Appendix E: User Guide to Downloading and Using Six-Year Review 4 and Related Data from EPA's Website

This appendix includes a user guide for downloading and using the Six-Year Review 4 (SYR 4) and related data from EPA's website. This document is also posted online with the data. In addition, instructions on importing the SYR 4 datasets are included in this Appendix (see Section 10). The data dictionary for all datasets is also included in Appendix C above.

Several of the contaminant occurrence datasets that are posted online were not analyzed as part of the SYR 4 effort. These contaminants were not subject to detailed review in SYR 4 due to recent, ongoing, or pending regulatory action (e.g., lead, copper, DBPs). These datasets passed the same QA procedures as those analyzed in SYR 4.

The data files are posted online in several zip files. Each zip file includes text files for multiple contaminants/parameters. The number of records and contaminants/parameters included in each file varies. The user may want to compare their counts of records downloaded for each contaminant of interest to the table of records provided in this user guide's exhibits to ensure that all of the records were correctly downloaded and imported. Note that these record counts reflect the data after the QA/QC process. For a list of data elements included in the data posted online, refer to Exhibit E.1.

The remainder of this document is organized as follows:

- Section 1: Background Information on Six-Year Review 4 Data Records
- Section 2: SYR 4 Data Records Posted for Phase Chemicals, Lead, Copper and Radionuclides
- Section 3: SYR 4 Data Records Posted for Disinfection Byproducts
- Section 4: SYR 4 Data Records Posted for Disinfection Byproducts Related Parameters
- Section 5: SYR 4 Data Records Posted for Microbial Contaminants, Microbial Related Parameters, and Disinfectant Residuals
- Section 6: SYR 4 Data Records Posted for the Aircraft Drinking Water Rule (ADWR)
- Section 7: Additional Data Collected under SYR 4 ICR
- Section 8: SYR 4 Data Records Posted for Treatment
- Section 9: SYR 4 Data Considerations
- Section 10: Instructions on Importing SYR 4 Datasets
    - 10A: Downloading Data Files
    - 10B: Importing Data into Microsoft Excel
    - 10C: Importing Data into R
    - 10D: Importing Data into Microsoft Access

Section 1: Background Information on SYR 4 Data Records

To support the national contaminant occurrence and exposure assessments performed under the fourth Six-Year Review process (SYR 4), EPA collected compliance monitoring data and treatment technique information from public water systems (PWSs) for regulated drinking water contaminants. This analysis allows EPA to characterize the frequency of occurrence, the levels found, and the geographic distribution of contaminants and related data to help the agency determine if there may be a meaningful opportunity to improve public health protection. EPA conducted a voluntary data request from states, primacy agencies, territories, and tribes (referred to as "States" throughout the remainder of this Appendix) to obtain compliance monitoring data and treatment technique information necessary to analyze national contaminant occurrence in support of SYR 4. This data request was conducted through the Information Collection Request (ICR) process. EPA requested States to submit their Safe Drinking Water Act (SDWA) compliance monitoring data and treatment technique information collected between January 2012 and December 2019. For more information on the process undertaken to request the voluntary submission of compliance monitoring data and treatment technique information by the States, see the fourth Six-Year Review ICR (84 FR 58381, USEPA, 2019).

Through extensive data management efforts, quality assurance evaluations, and communications and consultations with State's data management staff, EPA established a single contaminant occurrence dataset that consists of compliance monitoring data and treatment technique information from 59 out of 66 jurisdictions (46 states plus Washington, D.C., American Samoa, Navajo Nation, Commonwealth of the Northern Mariana Islands, and other tribes). This dataset is referred to as the National Compliance Monitoring ICR dataset for the fourth Six-Year Review (SYR 4 ICR dataset). The 59 States that provided data for the SYR 4 ICR dataset comprise 88 percent of all PWSs and 92 percent of the total population served by PWSs nationally, and are geographically representative of PWSs nationwide. The SYR 4 ICR dataset was used to estimate a variety of occurrence measures to characterize the national occurrence of regulated contaminants in public water systems to support the Six-Year Review process.

EPA received compliance monitoring data and treatment technique information from both SDWIS/State and non-SDWIS/State users. For States that use SDWIS/State, EPA developed a tool, available upon request from States, to extract the requested data identified in the SYR 4 ICR from a SDWIS/State database. In all, 46 states and 13 other jurisdictions provided compliance monitoring data that included parametric records. Thirty-five states, Washington D.C, and six regional tribal entities used the extraction tool to extract all or some of their data. The 17 States not using SDWIS/State submitted their compliance monitoring data and treatment technique "as is," resulting in a variety of formats, including dBase, Excel, XML, Access, and comma-delimited. With the exception of two States whose data were downloaded from their publicly available website (California and Florida), all States submitted their data online via EPA's Central Data Exchange. All data were conformed to a similar format with consistent units of measurement for consistency. For more details about the collection and formatting of SYR 4 ICR data, see the main chapters of this document.

EPA conducted a quality assurance and control evaluation of these data submitted by States and assembled these data into the SYR 4 ICR database, which includes more than 83 million records

from approximately 142,000 public water systems, serving approximately 303 million people nationally. The dataset includes the results of all compliance monitoring data (i.e., all sample analytical detections and non-detections) from January 2012 to December 2019 for regulated chemical phase contaminants, radionuclides, disinfectants and disinfection byproducts (D/DBPs), DBP precursors, microbial contaminants/indicators, disinfectant residuals, and other related data including treatment information. As noted in the main chapters, only the data that passed the QA/QC process are posted online.

Exhibit E.1, Six-Year Review 4 Data Field Names and Definitions, contains a list of the data elements, column names and a brief description of the data for each data element included in the SYR 4 ICR data text files.

Exhibit E.1: Six-Year Review 4 Data Field Names and Definitions

Data Element	Column Name	Description
Contaminant Identification Code	ANALYTE_CODE	4-digit Safe Drinking Water Information System (SDWIS) contaminant identification number for which the sample is being analyzed.
Contaminant Name	ANALYTE_NAME	Common name of contaminant for which the sample is being analyzed.
Primacy Code	PRIMACY_CODE	2- digit code identifying the primacy agency (i.e., State) for the water system.
State Code	STATE_CODE	2-digit code identifying the U.S. state or territory in which the water system is located.
Public Water System Identification Number (PWSID)	PWSID	The code used to identify each PWS. The code begins with the standard 2- character postal state abbreviation or region code; the remaining 7 numbers are unique to each PWS in the State.
System Name	SYSTEM_NAME	Name of the PWS.
Federal Public Water System Type Code	SYSTEM_TYPE	A code to identify whether a system is: • Community Water System (C); • Non-Transient Non-Community Water System (NTNC); or • Transient Non-Community Water System (NC).
Retail Population served	RETAIL_POPULATIO N_SERVED	Retail population served by a system.
Adjusted Total Population-served	ADJUSTED_TOTAL_ POPULATION_ SERVED	Adjusted total population served (retail plus adjusted wholesale population served as not to double-count buyer systems that purchase from multiple seller systems).
Source Water Type	SOURCE_WATER_ TYPE	Type of water at the source. Source water type can be: • Ground water (GW); • Surface water (SW); • Purchased Surface Water (SWP); • Purchased Ground Water (GWP); • Ground Water Under Direct Influence of Surface Water (GU); or • Purchased Ground Water Under Direct Influence of Surface Water (GUP).
Facility Identification Code	WATER_FACILITY_ID	A unique identifier for each water system facility.
Water Facility Type	WATER_FACILITY_ TYPE	Type of water system facility: • CC = Consecutive Connection; • CH = Common Headers; • CW = Clear Well; • DS = Distribution System; • IG = Infiltration Gallery; • IN = Intake;

Data Element	Column Name	Description
		• OT = Other; • PC = Pressure Control; • PF = Pumping Facility; • RS = Reservoir; • SI = Surface Impoundment; • SP = Spring; • SS = Sampling Station; • ST = Storage; • TM = Transmission Main (Manifold); • TP = Treatment Plant; • WH = Well Head; • WL = Well; or • XX = unknown.
Sampling Point Identification Code	SAMPLING_POINT_ID	A unique identifier for each sampling point location.
Sampling Point Type	SAMPLING_POINT_TYPE	Location type of a sampling point: • DS = Distribution System; • EP = Entry point; • FC = First Customer; • FN = Finished Water Source; • LD = Lowest Disinfectant Residual; • MD = Midpoint in the Distribution System; • MR = Point of Maximum Residence; • PC = Process Control; • RW = Raw Water Source; • SR = Source Water Point; • UP = Unit Process; or • WS = Water System Facility Point
Source Type Code	SOURCE_TYPE_CODE	Type of water source, based on whether treatment has taken place. Source type can be: • Finished (FN); • Raw (RW); or • Unknown (null or X).
Sample Type Code	SAMPLE_TYPE_CODE	Type of sample: • CO = Confirmation; • MR = Maximum Residence Time; • RP = Repeat; • RT = Routine; • ST = Split; • MS = Matrix spike; • TG = Triggered; or • FB = Field Blank.
Laboratory Assigned Identification Number	LABORATORY_ASSIGNED_ID	Unique lab identification, used to link up the total coliform positive (TC+) and *E. coli* / fecal coliform samples.
Six-Year ID	SIX_YEAR_ID	Unique identifier for each analytical result.
Sample Identification Number	SAMPLE_ID	Identifier assigned by State or the laboratory that uniquely identifies a sample.
Sample Collection Date	SAMPLE_COLLECTION_DATE	Date the sample was collected, including month, day, and year.
Detection Limit Value	DETECTION_LIMIT_VALUE	Limit below which the specific lab indicated they could not reliably measure results for a contaminant with the methods and procedures used by the lab.
Detection Limit Unit	DETECTION_LIMIT_UNIT	Units of the detection limit value.
Detection Limit Code	DETECTION_LIMIT_CODE	Indicates the type of Detection Limit reported in the Detection Limit Value column (e.g., the Minimum Reporting Level, Laboratory Reporting Level, etc.)
Sample Analytical Result - Sign	DETECT	The sign indicates whether the sample analytical result was: • (0) "less than" means the contaminant was not detected or was

Data Element	Column Name	Description
		detected at a level "less than" the MRL. • (1) "equal to" means the contaminant was detected at a level "equal to" the value reported in "Sample Analytical Result - Value."
Sample Analytical Result - Value	VALUE	For detections, this field is equal to the actual numeric (decimal) value of the analysis for the chemical result; for non-detections, this field is blank.
Sample Analytical Result - Unit of Measure	UNIT	Unit of measurement for the analytical results reported (usually expressed in either µg/L or mg/L for chemicals; or pCi/L for radionuclides).
Presence Indicator Code	PRESENCE_ INDICATOR_CODE	Indication of whether results of an analysis were positive or negative for TC, EC and FC. • P = Presence • A = Absence.
Residual Field Free Chlorine	RESIDUAL_FIELD_ FREE_CHLORINE_M G_L	Amount of free chlorine residual (in mg/L) found in the water after disinfectant has been applied. These concentrations were measured in the field at the same time and location as coliform samples (TC-EC-FC samples).
Residual Field Total Chlorine	RESIDUAL_FIELD_ TOTAL_CHLORINE_ MG_L	Amount of total chlorine residual (in mg/L) found in the water after disinfectant has been applied. These concentrations were measured in the field at the same time and location as coliform samples (TC-EC-FC samples).

Section 2: SYR 4 Data Records Posted for Phase Chemicals, Lead, Copper and Radionuclides

Exhibit E.2 provides a count of States, total number of sample records and systems for each phase chemical, lead, copper, and radionuclide collected for SYR 4. Contaminant occurrence data are grouped into zip files, which are indicated in the final column of Exhibit E2.

Exhibit E.2: Number of Six-Year Review 4 Data Records for Phase Chemicals, Lead, Copper, and Radionuclides and Zip Filename(s)

Contaminant	Analyte ID	Number of States	Total Number of Sample Records	Total Number of Systems	Zip Filename
Phase Chemicals					
1,1,1-Trichloroethane	2981	58	491,411	52,207	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
1,1,2-Trichloroethane	2985	58	482,294	52,200	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
1,1-Dichloroethylene	2977	58	508,764	52,206	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
1,2,4-Trichlorobenzene	2378	58	480,039	52,201	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
1,2-Dibromo-3-chloropropane	2931	57	244,298	37,153	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
1,2-Dichloroethane	2980	58	493,514	52,209	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
1,2-Dichloropropane	2983	58	481,065	52,197	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
2,3,7,8-TCDD	2063	42	38,934	6,222	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
2,4,5-TP	2110	58	187,025	40,954	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
2,4-D	2105	58	191,658	41,519	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
Alachlor	2051	58	215,965	42,822	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
Antimony, Total	1074	59	230,942	51,063	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
Arsenic	1005	59	452,852	52,505	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
Asbestos	1094	48	24,124	13,772	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
Atrazine	2050	58	225,827	43,763	SYR4_PhaseChem_1 (1,1,1-Trichloroethane to Atrazine).zip
Barium	1010	59	232,216	52,488	SYR4_PhaseChem_2 (Barium to Cyanide).zip
Benzene	2990	58	487,631	52,207	SYR4_PhaseChem_2 (Barium to Cyanide).zip
Benzo(A)pyrene	2306	58	190,003	35,877	SYR4_PhaseChem_2 (Barium to Cyanide).zip
Beryllium, Total	1075	59	229,630	50,225	SYR4_PhaseChem_2 (Barium to Cyanide).zip
BHC-Gamma	2010	58	195,775	38,843	SYR4_PhaseChem_2 (Barium to Cyanide).zip

Contaminant	Analyte ID	Number of States	Total Number of Sample Records	Total Number of Systems	Zip Filename
Cadmium	1015	59	230,098	50,989	SYR4_PhaseChem_2 (Barium to Cyanide).zip
Carbofuran	2046	58	176,608	37,375	SYR4_PhaseChem_2 (Barium to Cyanide).zip
Carbon Tetrachloride	2982	58	510,599	52,205	SYR4_PhaseChem_2 (Barium to Cyanide).zip
Chlordane	2959	58	189,512	38,310	SYR4_PhaseChem_2 (Barium to Cyanide).zip
Chlorobenzene	2989	58	479,909	52,184	SYR4_PhaseChem_2 (Barium to Cyanide).zip
Chromium	1020	59	238,413	51,357	SYR4_PhaseChem_2 (Barium to Cyanide).zip
cis-1,2-Dichloroethylene	2380	58	495,228	52,210	SYR4_PhaseChem_2 (Barium to Cyanide).zip
Cyanide	1024	57	163,373	38,760	SYR4_PhaseChem_2 (Barium to Cyanide).zip
Dalapon	2031	58	232,471	40,062	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Di(2-Ethylhexyl) Adipate	2035	58	192,447	36,369	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Di(2-Ethylhexyl) Phthalate	2039	58	202,419	36,486	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Dichloromethane	2964	58	487,166	52,222	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Dinoseb	2041	58	186,403	40,854	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Diquat	2032	54	110,637	22,215	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Endothall	2033	51	98,015	18,624	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Endrin	2005	58	192,869	38,483	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Ethylbenzene	2992	58	487,555	52,200	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Ethylene Dibromide	2946	57	243,161	38,371	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Fluoride[1]	1025	59	435,466	52,202	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Glyphosate	2034	55	105,084	21,744	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Heptachlor	2065	58	193,927	38,640	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Heptachlor Epoxide	2067	58	193,623	38,638	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Hexachlorobenzene	2274	58	195,150	38,311	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Hexachlorocyclopentadiene	2042	58	196,236	38,471	SYR4_PhaseChem_3 (Dalapon to Hexachlorocyclopentadiene).zip
Mercury	1035	59	226,418	50,990	SYR4_PhaseChem_4 (Hybrid Nitrate to Nitrate).zip
Methoxychlor	2015	58	196,131	38,834	SYR4_PhaseChem_4 (Hybrid Nitrate to Nitrate).zip
Nitrate	1040	59	1,404,609	105,202	SYR4_PhaseChem_4 (Hybrid Nitrate to Nitrate).zip
Nitrate (Hybrid)[2]	1040/ 1038	59	1,635,300	127,904	SYR4_PhaseChem_4 (Hybrid Nitrate to Nitrate).zip

Contaminant	Analyte ID	Number of States	Total Number of Sample Records	Total Number of Systems	Zip Filename
Nitrite	1041	59	512,234	73,442	SYR4_PhaseChem_5 (Nitrate-Nitrite to Total Polychlorinated Biphenyls (PCB)
Nitrate-Nitrite	1038	51	561,314	76,530	SYR4_PhaseChem_5 (Nitrate-Nitrite to Total Polychlorinated Biphenyls (PCB)
o-Dichlorobenzene	2968	58	480,075	52,200	SYR4_PhaseChem_5 (Nitrate-Nitrite to Total Polychlorinated Biphenyls (PCB)
Oxamyl	2036	58	175,728	37,235	SYR4_PhaseChem_5 (Nitrate-Nitrite to Total Polychlorinated Biphenyls (PCB)
p-Dichlorobenzene	2969	58	480,247	52,203	SYR4_PhaseChem_5 (Nitrate-Nitrite to Total Polychlorinated Biphenyls (PCB)
Pentachlorophenol	2326	58	201,636	41,094	SYR4_PhaseChem_5 (Nitrate-Nitrite to Total Polychlorinated Biphenyls (PCB)
Picloram	2040	58	188,833	41,375	SYR4_PhaseChem_5 (Nitrate-Nitrite to Total Polychlorinated Biphenyls (PCB)
Selenium	1045	59	232,598	51,317	SYR4_PhaseChem_5 (Nitrate-Nitrite to Total Polychlorinated Biphenyls (PCB)
Simazine	2037	58	220,013	43,211	SYR4_PhaseChem_5 (Nitrate-Nitrite to Total Polychlorinated Biphenyls (PCB)
Styrene	2996	58	479,601	52,187	SYR4_PhaseChem_5 (Nitrate-Nitrite to Total Polychlorinated Biphenyls (PCB)
Tetrachloroethylene	2987	58	544,460	52,210	SYR4_PhaseChem_5 (Nitrate-Nitrite to Total Polychlorinated Biphenyls (PCB)
Thallium, Total	1085	59	229,685	51,007	SYR4_PhaseChem_5 (Nitrate-Nitrite to Total Polychlorinated Biphenyls (PCB)
Toluene	2991	58	488,192	52,348	SYR4_PhaseChem_5 (Nitrate-Nitrite to Total Polychlorinated Biphenyls (PCB)
Total Polychlorinated Biphenyls (PCB)	2383	49	116,454	23,262	SYR4_PhaseChem_5 (Nitrate-Nitrite to Total Polychlorinated Biphenyls (PCB)
Toxaphene	2020	58	183,765	37,419	SYR4_PhaseChem_6 (Toxaphene to Xylenes, total).zip
trans-1,2-Dichloroethylene	2979	58	488,716	52,194	SYR4_PhaseChem_6 (Toxaphene to Xylenes, total).zip
Trichloroethylene	2984	58	540,777	52,222	SYR4_PhaseChem_6 (Toxaphene to Xylenes, total).zip
Vinyl Chloride	2976	58	482,672	52,021	SYR4_PhaseChem_6 (Toxaphene to Xylenes, total).zip
Xylenes, Total	2955	56	412,436	46,720	SYR4_PhaseChem_6 (Toxaphene to Xylenes, total).zip
Lead and Copper					
Lead	1030	54	1,552,995	53,058	SYR4_PhaseChem_4 (Hybrid nitrate to nitrate).zip

Contaminant	Analyte ID	Number of States	Total Number of Sample Records	Total Number of Systems	Zip Filename
Copper	1022	55	1,579,728	54,224	SYR4_PhaseChem_2 (Barium to Cyanide).zip
Radionuclides					
Gross Alpha, Excl. Radon & U	4000	55	64,413	16,925	SYR4_Rads.zip
Gross Beta Particle Activity	4100	50	48,520	11,261	SYR4_Rads.zip
Combined Radium (-226 & -228)	4010	53	86,594	21,972	SYR4_Rads.zip
Combined Uranium	4006	55	97,663	18,491	SYR4_Rads.zip

[1] Includes records that passed the QA/QC procedures described in this document. See USEPA (2024c) for additional information on procedures conducted for the occurrence analysis.
[2] Includes all sampling results for nitrate and sampling results for total nitrate plus nitrite for systems for which there were no SYR 4 nitrate (only) data.

Section 3: SYR 4 Data Records Posted for Disinfection Byproducts

Exhibit E.3 provides a count of States, total number of sample records and systems for each regulated disinfection byproduct collected for SYR 4, and the zip files names that the data files can be located. These data records were not analyzed under SYR 4 because of the ongoing considerations of potential revisions of the Stage 1 and Stage 2 DBP Rules.

Exhibit E.3: Number of Six-Year Review 4 Data Records for Disinfection Byproducts and Zip filename(s)

Contaminant	Analyte ID	Number of States	Total Number of Sample Records	Total Number of Systems	Zip Filename
Disinfection Byproducts - Full Datasets					
Total Trihalomethanes (TTHM)	2950	57	1,089,557	46,297	SYR4_THMs.zip
Dibromochloromethane	2944	46	981,059	47,172	SYR4_THMs.zip
Bromoform	2942	46	976,412	47,129	SYR4_THMs.zip
Chloroform	2941	46	981,289	47,403	SYR4_THMs.zip
Bromodichloromethane	2943	46	977,561	47,196	SYR4_THMs.zip
Haloacetic Acids (HAA5)	2456	57	1,005,235	43,577	SYR4_HAAs.zip
Dibromoacetic Acid	2454	44	720,986	36,121	SYR4_HAAs.zip
Dichloroacetic Acid	2451	44	721,017	36,134	SYR4_HAAs.zip
Monochloroacetic Acid	2450	44	720,474	36,113	SYR4_HAAs.zip
Trichloroacetic Acid	2452	44	720,706	36,125	SYR4_HAAs.zip
Monobromoacetic Acid	2453	44	720,595	36,095	SYR4_HAAs.zip
Bromate	1011	38	23,298	444	SYR4_Bromate_Chlorite.zip
Chlorite	1009	33	87,995	514	SYR4_Bromate_Chlorite.zip

Note: The speciation data is higher for TTHM than HAA5 (90+% vs 70+%). There were two more States that provided speciated THM results as compared to speciated HAA results. About 11,000 systems had speciated THM data but not speciated HAA data. There are only about 200 systems with speciated HAA data but no speciated THM data. In addition, the number of PWSs providing speciated TTHM data is higher than number of PWSs providing TTHM. There are about 8,000 systems that have data for the speciated THMs but not TTHM whereas there are only about 7,000 systems with data for TTHM but not the speciated THMs.

Section 4: SYR 4 Data Records Posted for Disinfection Byproduct Related Parameters

This DBP-related data includes total organic carbon (TOC), alkalinity, pH, dissolved organic carbon (DOC), specific UV-absorbance (SUVA), and UV-absorbance. Full datasets for TOC and alkalinity (i.e., text files including all individual sample analytical results for TOC and alkalinity) are included. In addition to the full datasets for TOC and alkalinity, a paired TOC-alkalinity dataset was created that included, for each treatment plant (listed in Exhibit E.2 as a water system facility), the average monthly concentrations of TOC and alkalinity in raw water paired with the corresponding average finished water concentration of TOC. The paired TOC-alkalinity dataset was created to evaluate the percent removal of TOC using the SYR 4 data and joined the average monthly TOC concentration with the average monthly alkalinity concentration for individual water system facilities when possible. This paired dataset is directly related to the treatment technique requirements for TOC removal under the Stage 1 DBPR. EPA produced these datasets to support the ongoing considerations of potential revisions of the Stage 1 DBP Rule (85 FR 61680, USEPA, 2020). EPA did not analyze these data records under the SYR 4 effort. Historical efforts to evaluate the paired TOC-alkalinity data are described in *Six-Year Review 3 Technical Support Document for Disinfectants/Disinfection Byproducts Rules* (USEPA, 2016).

Exhibit E.4 provides a count of States, total number of sample records and systems for TOC (raw and finished), alkalinity, pH, DOC, SUVA, and UV-absorbance. The count of systems for raw and finished TOC samples are counted separately, so systems with samples in both categories are counted twice. The "full" TOC dataset contains only the raw/finished water designations from the original data provided by the State (see SOURCE_TYPE_CODE). However, for the "paired" TOC-alkalinity dataset, EPA applies the following logic to assign raw/finished water designations to records that were missing it. Raw samples are identified as samples taken at source water sampling points. Records are marked as raw if SOURCE_TYPE_CODE equals "RW" or SOURCE_TYPE_CODE is NULL but the water system facility type code equals "IG", "IN", "RS", "SP", "WL", or "CC". Records are marked as finished if SOURCE_TYPE_CODE equals "FN" or SOURCE_TYPE_CODE is NULL but the water facility type code equals "CW", "DS", "PF", "ST", "TM", "TP". Exhibit E.5 contains the list of data elements, column names, and a brief description of the data for each data element included in the "paired" TOC-alkalinity dataset. For a list of data elements included in the "full" TOC, alkalinity, and pH datasets, refer to Exhibit E.1.

Exhibit E.4: Number of Six-Year Review 4 Data Records for TOC, Alkalinity, pH, DOC, SUVA, and UV-absorbance and Zip Filename(s)

Contaminant	Analyte ID	Number of States	Total Number of Sample Records	Total Number of Systems	Zip Filename
Disinfection Byproduct Related Parameters – Full Datasets					
Total Organic Carbon (TOC)	2920	49	440,197	3,156	SYR4_DBP_Related Parameters.zip
Raw TOC	2920	42	188,358	2,494	SYR4_DBP_Related

Contaminant	Analyte ID	Number of States	Total Number of Sample Records	Total Number of Systems	Zip Filename
					Parameters.zip
Finished TOC	2920	38	155,558	1,999	SYR4_DBP_Related Parameters.zip
Alkalinity	1927	51	429,397	18,140	SYR4_DBP_Related Parameters.zip
pH	1925	52	632,821	28,660	SYR4_DBP_Related Parameters.zip
SUVA	2923	2	8,026	59	SYR4_DBP_Related Parameters.zip
UV-254	2922	3	6,061	60	SYR4_DBP_Related Parameters.zip
DOC	2919	3	5,908	76	SYR4_DBP_Related Parameters.zip
Disinfection Byproduct Related Parameters – Paired Dataset					
Paired TOC-alkalinity record	N/A	33	92,666	1,192	SYR4_DBP_Related Parameters.zip

Exhibit E.5 Paired TOC-Alkalinity Dataset Field Names and Definitions

Data Element	Column Name	Description
Public Water System Identification Number (PWSID)	NUMBER0	The code used to identify each PWS. The code begins with the standard 2- character postal state abbreviation or region code; the remaining 7 numbers are unique to each PWS in the state.
Sample Collection Date (Month)	Month	Month (1 through 12).
Sample Collection Date (Year)	Year	Year (2012 through 2019).
Retail Population-served	Population Served	Retail population served by the water system.
Federal Public Water System Type Code	System Type	Water system type according to federal requirements. C = Community water system NTNC = Non-transient non-community water system
Source Water Type	Source Water Type	Primary water source for the water system. GU = Ground water Under Direct Influence of Surface Water GW = Ground Water GWP = Purchased Ground Water SW = Surface Water SWP = Purchased Surface Water
Facility Identification Code	Water Facility ID	Unique identifier for each water system facility.
State Facility Identification Code	State Facility ID	Identifier for each water system facility that is unique within a particular state.
State Assigned Identification Code	State Assigned ID	A state-assigned value which identifies the water system facility.
Raw water TOC average concentration	Avg Of Raw TOC (mg/L)	Monthly average (in mg/L) total organic carbon (TOC) concentration in raw water.
Raw water alkalinity average concentration	Avg Of Raw Alkalinity (mg/L)	Monthly average (in mg/L) alkalinity concentration in raw water.
Finished water TOC average concentration	Avg Of Finished TOC (mg/L)	Monthly average (in mg/L) total organic carbon (TOC) concentration in finished water.

Section 5: SYR 4 Data Records Posted for Microbial Contaminants, Microbial Related Parameters, and Disinfectant Residuals

Data for three microbial contaminants (total coliforms (TC), *Escherichia coli* (EC), and fecal coliform (FC)) were collected from 2012 to 2019 for SYR 4. The TC datasets are separated into individual files by each year of data collected because of the large volume of data collected. Unlike the TC records which are provided separately by year, the EC and FC are contained in one file. The EC dataset is one large file intended for use in Access or R. Systems are required under the Surface Water Treatment Rule to monitor for disinfectant residuals at the same time and locations as TC under TCR/RTCR. Most States submitted data from systems that included free and total residual chlorine results paired with TC records. However, some States provided the residual monitoring data in separate datafiles or did not submit that information under the SYR 4 ICR.

Exhibit E.6 provides a count of States, total number of sample records and systems for TC, EC, FC, and records of disinfectant residuals. Exhibit E.6 also shows that some States submitted chlorine residual monitoring results separately under different analyte codes (e.g., Chlorine (Analyte ID 0999), Residual Chlorine (Analyte ID 1012), and Free Residual Chlorine (Analyte ID 1013)). To maximize the number of paired total coliform and chlorine residual records, EPA took additional steps to add records from States reporting residual data records separately (see Section 5.5.2 of the main text for details on pairing and the analytes used). The "full" datafiles in Exhibit E.6 contain these paired records as well as records for systems with reported microbial indicator presence and absence but no associated disinfection residual information.

To assist the user, EPA produced the "paired" TC, EC, and FC datafiles (Exhibit E.6), which contain only the records for systems in the "full" versions of those datasets that include paired residual information. The "paired" datafiles were not analyzed under SYR 4 because of the ongoing considerations of potential revisions of the Surface Water Treatment Rules.

Note that the TC, EC, and FC datasets contain the monitoring records under TCR/RTCR for systems with all source water types. The HPC, *Giardia*, disinfectant residual, and paired TC/EC/FC disinfectant residual files contain the monitoring records under the SWTRs. See Exhibit E.1 for a description of field names.

Exhibit E.6: Number of Six-Year Review 4 Data Records for Microbial Contaminants, Microbial Related Parameters, and Disinfectant Residuals and Zip Filename(s)

Contaminant	Analyte ID	Number of States/ Entities with Data	Total Number of Sample Records	Total Number of Systems	Zip Filename
Microbial Contaminants and Disinfectant Residual – Full Datasets					
Total Coliform (2012)	3100	54	2,349,687	102,423	SYR4_TC.zip

Contaminant	Analyte ID	Number of States/ Entities with Data	Total Number of Sample Records	Total Number of Systems	Zip Filename
Total Coliform (2013)	3100	54	2,398,740	102,713	SYR4_TC.zip
Total Coliform (2014)	3100	56	2,521,212	105,515	SYR4_TC.zip
Total Coliform (2015)	3100	56	2,513,937	104,532	SYR4_TC.zip
Total Coliform (2016)	3100	57	2,656,932	113,099	SYR4_TC.zip
Total Coliform (2017)	3100	57	2,780,743	114,328	SYR4_TC.zip
Total Coliform (2018)	3100	57	2,849,385	114,954	SYR4_TC.zip
Total Coliform (2019)	3100	57	2,675,476	111,385	SYR4_TC.zip
E. coli (EC)	3014	57	7,175,363	93,728	SYR4_EC_FC_HPC_Giardia.zip
E. coli (EC) In Raw Water[1]	3014	43	65,805	19,515	SYR4_EC_FC_HPC_Giardia.zip
E. coli (EC) In Distribution Systems[2]	3014	49	6,346,973	90,607	SYR4_EC_FC_HPC_Giardia.zip
E. coli (EC) In Unknown Sampling Location[3]	3014	54	762,585	24,486	SYR4_EC_FC_HPC_Giardia.zip
Fecal Coliform (FC)	3013	40	16,818	1,835	SYR4_EC_FC_HPC_Giardia.zip
Coliphage	3028	2	3	3	SYR4_EC_FC_HPC_Giardia.zip
Enterococci	3002	3	8	4	SYR4_EC_FC_HPC_Giardia.zip
Cryptosporidium	3015	29	19,542	740	SYR4_EC_FC_HPC_Giardia.zip
Heterotrophic Bacteria (HPC)	3001	16	135,081	595	SYR4_EC_FC_HPC_Giardia.zip
Giardia Lamblia	3008	15	4,628	229	SYR4_EC_FC_HPC_Giardia.zip
Legionella		0	0	0	N/A
Chlorine[4]	0999	19	6,100,133	4,438	SYR4_Disinfectant_ Residuals.zip
Total Chlorine[4]	1000	1	125,788	741	SYR4_Disinfectant_ Residuals.zip
Chloramine[4]	1006	9	78,664	198	SYR4_Disinfectant_ Residuals.zip
Residual Chlorine[4]	1012	4	179,599	572	SYR4_Disinfectant_ Residuals.zip
Free Residual Chlorine[4]	1013	3	2,000,997	4,044	SYR4_Disinfectant_ Residuals.zip
Chlorine Dioxide	1008	9	12,752	28	SYR4_Disinfectant_ Residuals.zip
Microbes and Associated Disinfectant Residuals – Paired Datasets[5]					
E. coli (EC) with Associated Disinfectant Residuals	3014	49	3,079,032	28,091	SYR4_Paired Microbes_DR.zip

Contaminant	Analyte ID	Number of States/ Entities with Data	Total Number of Sample Records	Total Number of Systems	Zip Filename
Fecal Coliform (FC) with Associated Disinfectant Residuals	3013	24	5,966	534	SYR4_Paired Microbes_DR.zip
Total Coliform (TC) paired with Associated Disinfectant Residuals (2012)	3100	43	1,165,209	30,950	SYR4_Paired Microbes_DR.zip
Total Coliform (TC) paired with Associated Disinfectant Residuals (2013)	3100	44	1,173,926	31,132	SYR4_Paired Microbes_DR.zip
Total Coliform (TC) paired with Associated Disinfectant Residuals (2014)	3100	46	1,218,722	31,865	SYR4_Paired Microbes_DR.zip
Total Coliform (TC) paired with Associated Disinfectant Residuals (2015)	3100	47	1,241,995	31,880	SYR4_Paired Microbes_DR.zip
Total Coliform (TC) paired with Associated Disinfectant Residuals (2016)	3100	48	1,274,211	34,654	SYR4_Paired Microbes_DR.zip
Total Coliform (TC) paired with Associated Disinfectant Residuals (2017)	3100	50	1,331,868	37,217	SYR4_Paired Microbes_DR.zip
Total Coliform (TC) paired with Associated Disinfectant Residuals (2018)	3100	50	1,480,354	41,053	SYR4_Paired Microbes_DR.zip
Total Coliform (TC) paired with Associated Disinfectant Residuals (2019)	3100	50	1,498,050	38,029	SYR4_Paired Microbes_DR.zip

[1] Includes results with a sample type code of "TG" (i.e., triggered monitoring). Note that these record counts are subsets of the *E. coli* records included in the *E. coli* data set.
[2] Includes results not marked as triggered but had a sample point type of "DS", "FC", "FN", "LD", "MD", or "MR" or records with water facility type of "CC", "DS", "TP", or "TM" and sample point type of "WS" or null. Note that these record counts are subsets of the *E. coli* records included in the *E. coli* data set.
[3] Includes remaining *E. coli* results not identified as coming from raw water or the distribution system. Note that these record counts are subsets of the *E. coli* records included in the *E. coli* data set.
[4] Reported independently of the coliform sample results.
[5] Refer to Section 5.5.2 for more details on the paired disinfectant residual and total coliform records.

Section 6: SYR 4 Data Records Posted for Aircraft Drinking Water Rule (ADWR)

EPA downloaded compliance data from the Agency's Aircraft Reporting and Compliance System (ARCS) for the period from February 2011 to May 2021. This dataset includes aircraft compliance monitoring data for TC and EC for aircraft drinking water systems (Exhibit E.7). The Aircraft PWS Inventory file includes records for 8,627 unique aircraft drinking water systems. Details on the QA/QC procedure for this data can be found in Appendix D of this document. Note that the number of sample records presented below and included in the posted data reflect counts prior to the QA/QC procedures were applied for the SYR 4 analyses as presented in USEPA (2024b). After the QA/QC steps described in Appendix D are applied, there are 140,502 total coliform and 92,994 *E. coli* records remaining.

Exhibit E.7: Number of Aircraft Drinking Water Rule (ADWR) Data Records and Zip filename

Contaminant	Total Number of Sample Records	Total Number of Systems	Zip Filename
Aircraft PWS Sample by Air Carrier and Results			
Total Coliform	212,937	8,094	SYR4_ADWRCompliance Data.zip
E. coli[2]	93,011	7,091	SYR4_ADWR Compliance Data.zip

[1] The number of records presented here is greater than the number of rows of data downloaded from ARCS (70,979 at the time of download in support of the SYR 4 analysis) because it counts all samples within each row of data (i.e., Sample 1, Sample 2, and Sample 3). Note that Sample 3 is related to the ability to have third sample collected, which is not a requirement of ADWR and is not often used. Typically there is no data for Sample 3 fields.
[2] The count of *E. coli* records and systems is based on all *E. coli* samples listed as either "present" or "absent." It does not include samples listed as "not speciated" or "not analyzed."

Exhibit E.8, Data Dictionary Aircraft Drinking Water Rule (ADWR) Dataset, contains a list of the data elements, column names and a brief description of the data for each data element included in the ADWR data text files.

Exhibit E.8: Data Dictionary Aircraft Drinking Water Rule (ADWR) Dataset

Data Element	Column Name	Description
PWS Inventory		
Official FAA Corporate Name	FAA Corporate Name	The name of the air carrier or operator as registered with the FAA.
FAA Designator	FAA Designator	The four-character designator assigned to the air carrier by the FAA.
PWSID	PWS ID	The aircraft public water system identification number (PWS_ID) used by EPA to uniquely identify the aircraft public water system (PWS).
FAA Aircraft Registry No	FAARegistry No.	The number for the aircraft that is registered with the Federal Aviation Administration (FAA), commonly referred to as the N-number or tail number.
Aircraft Activity Status Code	Status	The activity status of the aircraft. It is selectable from the drop-down list. Permissible values are [Active] or [Inactive].
Routine Disinfection and Flushing Frequency	D&FFrequency	The frequency of routine disinfection and flushing scheduled for this aircraft.
Routine Sample Frequency	SamplingFrequency	The frequency of routine coliform sampling scheduled for this aircraft.
Aircraft Manufacturer	Manufacturer	The manufacturer of the aircraft.
Aircraft Model	Model	The manufacturer's model of the aircraft.
Seating Capacity	Seat Capacity	The number of passenger seats configured for the aircraft. It has a maximum value of 999.
Samples by Air Carrier		
Official FAA Corporate Name	FAA Corporate Name	The name of the air carrier or operator as registered with the FAA.

Data Element	Column Name	Description
FAA Aircraft Registry No	FAA Registry No.	The number for the aircraft that is registered with the Federal Aviation Administration (FAA), commonly referred to as the N-number or tail number.
PWSID	PWS ID	The aircraft public water system identification number (PWS_ID) used by EPA to uniquely identify the aircraft public water system (PWS).
Routine Sample Frequency	Routine Sample Frequency	The frequency of routine coliform sampling scheduled for this aircraft.
Sample Type	Sample Type	Indicates the type of individual sample: routine, repeat, follow-up, special.
Date and Time Collected	Sample Taken On	The date and time the sample was collected. When the galley and lavatory samples are collected on the same day, the date and time the first sample was collected is used. The required format is MM/DD/YYYY with time reported on a 24-hour clock as H:MI (e.g., 12/01/2014 15:00).
Date and Time Results Received	Samples Results On	The date and time the sample analysis results were received from the laboratory (e.g., phone message, USPS delivery date, office date and time stamp, e-mail receipt date and time). The required format is MM/DD/YYYY with time reported on a 24-hour clock as Hours:Minutes (e.g., 12/01/2014 15:00).
Sample Collection Location (Sample 1)	Sample1 Location	The location on the aircraft from where the first sample was collected. The options are [galley] or [lavatory].
Total Coliform Result (Sample 1)	Sample1 Total Coliform	The reported lab result that indicates the presence or absence of total coliform in the first sample analyzed. The drop-down list options are [Present] or [Absent].
E. coli Result (Sample 1)	Sample1 E.Coli	The lab analytical result that indicates the presence or absence of *E. coli* in the first sample analyzed. The drop-down list options are [Present] or [Absent] or [Did not speciate]. “Did not speciate” is used when the lab did not analyze a TC+ sample (or “present” sample result) for *E. coli*. Note: certified labs are required to analyze all TC+ samples for *E. coli*, but it is the carrier’s responsibility to make sure the lab completed the speciation.
Sample Collection Location (Sample 2)	Sample2 Location	The location on the aircraft from where the second sample was collected. The options are [galley] or [lavatory].
Total Coliform Result (Sample 2)	Sample2 Total Coliform	The reported lab result that indicates the presence or absence of total coliform in the second sample analyzed. The drop-down list options are [Present] or [Absent].
E. coli Result (Sample 2)	Sample2 *E. coli*	The lab analytical result that indicates the presence or absence of *E. coli* in the second sample analyzed. The drop-down list options are [Present] or [Absent] or [Did not speciate]. “Did not speciate” is used when the lab did not analyze a TC+ sample (or “present” sample result) for *E. coli*. Note: certified labs are required to analyze all TC+ samples for *E. coli*, but it is the carrier’s responsibility to make sure the lab completed the speciation.
Sample Collection Location (Sample 3)	Sample3 Location	The location on the aircraft from where the third sample was collected. The options are [galley] or [lavatory].
Total Coliform Result (Sample 3)	Sample3 Total Coliform	The reported lab result that indicates the presence or absence of total coliform in the third sample analyzed. The drop-down list options are [Present] or [Absent].
E. coli Result (Sample 3)	Sample3 *E. coli*	The lab analytical result that indicates the presence or absence of *E. coli* in the third sample analyzed. The drop-down list options are [Present] or [Absent] or [Did not speciate]. “Did not speciate” is used when the lab did not analyze a TC+ sample (or “present” sample result) for *E. coli*. Note: certified labs are required to analyze all TC+ samples for *E. coli*, but it is the carrier’s responsibility to make sure the lab completed the speciation.

Section 7: Additional Data Collected under SYR 4 ICR

Additional data relating to certain microbial rules were collected under the SYR 4 ICR request, including calculated compliance values and corrective actions information. Note that these data did not undergo the same quality assurance evaluations as the rest of the data.

Calculated Compliance Values

Exhibit E.9 provides a summary of the data elements included in the calculative compliance values related to *Cryptosporidium* binning information from SYR 4 ICR database. Exhibit E.10 provides a summary of the systems and states that provided SYR4 *Cryptosporidium* binning data.

Exhibit E.9: Data Dictionary of *Cryptosporidium* Binning Information Included as part of the Calculated Compliance Values Table (Filename: SYR4_CryptoBinning)

Data Element	Column Name	Description
Contaminant Identification Code	ANALYTE_CODE	4-digit Safe Drinking Water Information System (SDWIS) contaminant identification number for which the sample is being analyzed.
Contaminant Name	ANALYTE_NAME	Common name of contaminant for which the sample is being analyzed.
Public Water System Identification Number (PWSID)	PWSID	The code used to identify each PWS. The code begins with the standard 2- character postal State abbreviation or region code; the remaining 7 numbers are unique to each PWS in the State.
Facility Identification Code	WATER_FACILITY_ID	A unique identifier for each water system facility.
Compliance Period Begin Date	CP_PRD_BEGIN_ DT	Compliance Period Begin Date.
Compliance Period End Date	CP_PRD_END_DT	Compliance Period End Date.
Bin Number	BIN_NUMBER	The BIN assignment for the period of time covered by the average.

Exhibit E.10: Six-Year Review 4 Data Summary for Calculated Compliance Values Related to *Cryptosporidium* Binning Information

Number of States	Total Number of Sample Records	Total Number of Systems	Zip Filename
23	27,812	486	SYR4_CryptoBinning.zip

Corrective Actions

Exhibit E.11 provides a summary of the data elements included in the corrective actions table within the SYR 4 ICR database. Exhibit E.12 provides a summary of the corrective action data collected as part of SYR. Note, however, that EPA did not evaluate the specific types of corrective actions (e.g., those related to sanitary surveys) as part of SYR 4.

Exhibit E.11: Corrective Actions Data Dictionary (Filename: SYR4_CorrectiveActions)

Data Element	Column Name	Description
Corrective Action ID	CORACT_ID	Unique identifier for each corrective action.
Public Water System Identification Number (PWSID)	PWSID	The code used to identify each PWS. The code begins with the standard 2- character postal State abbreviation or region code; the remaining 7 numbers are unique to each PWS in the State.
State Code	STATE_CODE	State in which the system is located using the State's two letter abbreviation.
Date Issue Identified	DATE_ISSUE_IDENTIFIED	Date the corrective action was identified.
Schedule Type	SCHEDULE_TYPE	Type of schedule for the corrective action.
Schedule Description	SCHEDULE_DESCRIPTION	Schedule for the corrective action.
Corrective Action Category Code	CORACT_CAT_CODE	Category code for the corrective action.
Corrective Action Name	CORACT_NAME	Name of the corrective action.
Due Date	DUE_DATE	Due date for the required corrective action.
Achieved Date	ACHIEVED_DATE	The date that the water system achieved the corrective action required.

Exhibit E.12: Six-Year Review 4 Data Summary for Corrective Actions

Number of States	Total Number of Sample Records	Total Number of Systems	Zip Filename
41	69,821	15,984	SYR4_Corrective_Actions.zip

Section 8: Treatment Data

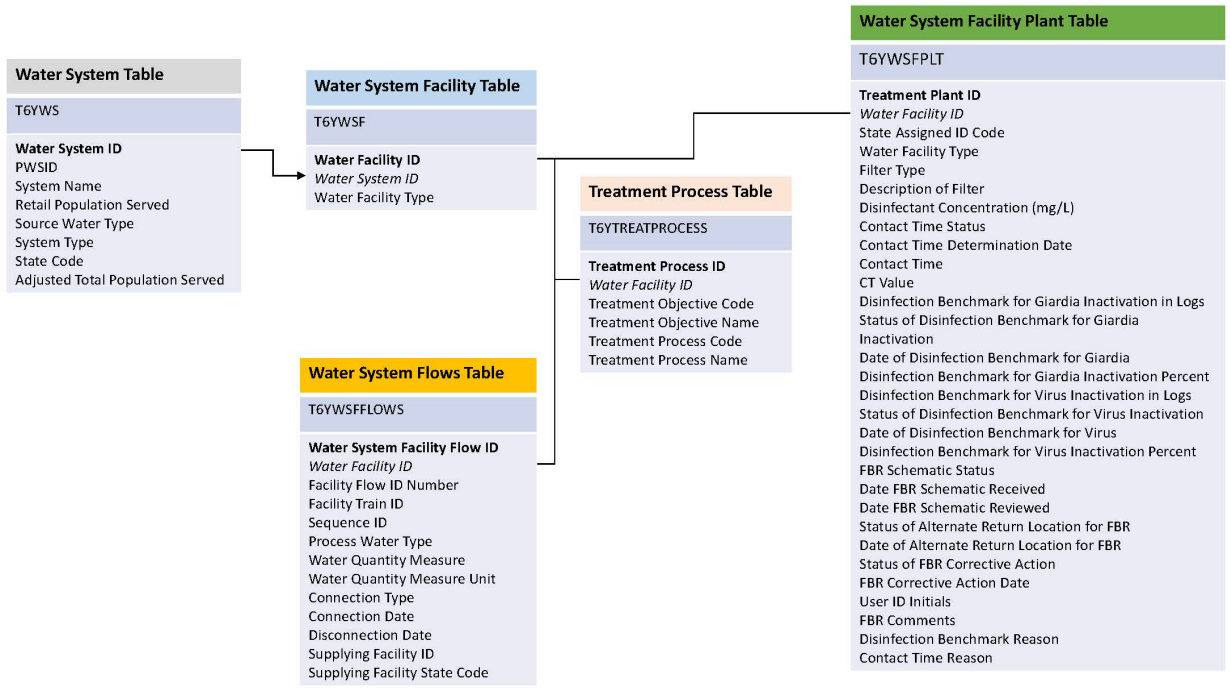
Exhibits E.13 and E.14 provide a comprehensive summary of the data elements included in the treatment information within the SYR 4 ICR database. EPA has posted these data online; however, it is important to note that the treatment information did not undergo the same quality assurance evaluations as the SYR 4 occurrence data. Exhibit E.13 identifies the data elements used in the treatment information tables and a description of each data element. However, the majority of these data elements are not populated in the SYR 4 ICR dataset. Exhibit E.14 represents the database relationships between tables in the SYR 4 ICR treatment database. This diagram shows how the treatment tables relate to one another. Bolded field names are primary keys, or unique fields, designated to identify all table records. Primary keys contain a unique number for each row of data. Italicized field names are foreign keys that serve as the link (i.e., connection) between two or more related tables. Relationships between key fields in different tables are illustrated by the lines connecting the tables.

Exhibit E.13: Treatment Data Dictionary (Filename: SYR4_Treatment)

Data Element	Description
Water system facility plant table (T6YWSFPLT)	
Treatment Plant ID	Unique identifier for each treatment plant water system facility record.
Water Facility ID	Identifier that relates each record to the unique record in the T6YWSF table.
State Assigned ID Code	A State-assigned value which identifies the treatment plant water system facility.
Water Facility Type	The value extracted from SDWIS/State will be “TP” (treatment plant). The values from non-SDWIS States include “TM” (transmission manifold) and “ST” (storage).
Filter Type	Unfiltered (UF), Conventional Filtration (CF), Direct Filtration (DF), Diatomaceous Earth (DE), Other (OT), and other permitted values that the System Administrator may add.
Description of Filter	A description of the filter.
Disinfectant Concentration (mg/L)	Disinfectant Concentration in mg/L.
Contact Time Status	Contact Time Status. Permitted values are: RQD – Required; NRQD - Not Required; REQT – Requested; RECV – Received; URVW - Under Review; RVWD – Reviewed; APVD – Approved; DTMD – Determined; DENY – Denied; RESB – Resubmitted.
Contact Time Determination Date	Date the Contact Time was determined.
Contact Time	Contact Time in minutes – the number of minutes the water was in contact with the disinfectant in order to be properly disinfected. The range of values is 0001 to 2400.
CT Value	CT value in mg x min/liter.
Disinfection Benchmark for *Giardia* Inactivation in Logs	The disinfection profile benchmark for *Giardia* inactivation in Logs.
Status of Disinfection Benchmark for *Giardia* Inactivation	The status of the disinfection profile benchmark for *Giardia* inactivation. See CONTACT_TIME_STAT for permitted values and description.
Date of Disinfection Benchmark for *Giardia*	The date the disinfection virus benchmark was determined.
Disinfection Benchmark for *Giardia* Inactivation Percent	The disinfection profile benchmark for *Giardia* inactivation percent.
Disinfection Benchmark for Virus Inactivation in Logs	The disinfection profile benchmark for virus inactivation in Logs.
Status of Disinfection Benchmark for Virus Inactivation	The status of the disinfection profile benchmark for Virus inactivation. See CONTACT_TIME_STAT for permitted values and description
Date of Disinfection Benchmark for Virus	The date the disinfection virus benchmark was determined.
Disinfection Benchmark for Virus Inactivation Percent	The disinfection profile benchmark for virus inactivation percent.
FBR Schematic Status	Under the Filter Backwash Rule, a water system is required to submit a schematic of this treatment plant to the primacy agency for review to demonstrate the percentage of filter backwash that is returned to the treatment plant influent. See CONTACT_TIME_STAT for permitted values and description.
Date FBR Schematic Received	Date primacy agency received treatment plant schematic to demonstrate the percentage of filter backwash that is returned to the treatment plant influent.
Date FBR Schematic Reviewed	Date primacy agency completes review of treatment plant schematic and determines the percentage of filter backwash that is returned to the treatment plant influent.
Status of Alternate Return Location for FBR	The status of a request from the water system to request an alternate location for return of the filter backwash
Date of Alternate Return Location for FBR	The date that the water system requested an alternate location for return of the filter backwash.
Status of FBR Corrective Action	The status of corrective action by the water system as required by the primacy agency after review of the schematic of the filter backwash flow in the treatment plant.

Data Element	Description
FBR Corrective Action Date	The date that the water system achieved the corrective action required for the filter backwash.
User ID Initials	The User ID of the person who created this record.
FBR Comments	A memo field into which a user may enter comments about the Filter Backwash Recycling Rule.
Disinfection Benchmark Reason	Text description associated with the Disinfection Benchmark Reason.
Contact Time Reason	Text description associated with the Contact Time.
Treatment process table (T6YTREATPROCESS)	
Treatment Process ID	Unique identifier for each treatment record.
Water Facility ID	Identifier that relates each record to the unique record in the T6YWSF table.
Treatment Objective Code	A coded value that categorizes the treatment objective.
Treatment Objective Name	The name of the treatment objective.
Treatment Process Code	A coded value that categorizes the treatment process.
Treatment Process Name	The name of the treatment process.
Water system flows table (T6YWSFFLOWS)	
Water System Facility Flow ID	Unique identifier for each water system facility flow record.
Water Facility ID	Identifier that relates each record to the unique record in the T6YWSF table.
Facility Flow ID Number	Identifier for each water system facility flow entry that is unique when combined with T6YWSFT6YWSF_ID.
Facility Train ID	This attribute identifies the water system facilities that are part of the same flow.
Sequence ID	This attribute identifies the order of the water system facilities in a specific flow.
Process Water Type	A system administrator controlled code of the type of water flowing between the facilities.
Water Quantity Measure	A value that represents the number of gallons of water purchased.
Water Quantity Measure Unit	A coded value which specifies the unit of measurement for the quantity of water purchased.
Connection Type	Categorizes the type of connection between the water system facilities.
Connection Date	The date of the connection of the water system facility to another water system facility.
Disconnection Date	The date of the disconnection of the water system facility from another water system facility.
Supplying Facility ID	Identifier for each supplying water system facility that is unique when combined with TINWSF0ST_CODE.
Supplying Facility State Code	Two-digit code that identifies the State that submitted data for the facility

Exhibit E.14: Treatment Data Diagram



Section 9: SYR 4 Data Considerations

The SYR 4 ICR data has undergone appropriate quality assurance evaluation and enough States provided compliance monitoring data and treatment technique information to be representative for national-scale analyses. EPA used the data in analytical activities informing decisions for SYR 4. The data include sufficient information for users to be able to reproduce the SYR 4 analyses. There are a few limitations of the final SYR 4 ICR dataset that should also be acknowledged. There may be different levels of completeness for different contaminants within the dataset. In some cases, the number of records per State ranged from less than one hundred records up to more than a million records for a given contaminant. States might not have submitted data for certain contaminants if they have monitoring waivers for the contaminant. States may grant waivers to PWSs to reduce monitoring frequencies, and it is possible that no samples were collected by systems during the SYR 4 period of review. Other States may have submitted data for these contaminants under the ICR; however, the data were not in a format compatible with the SYR 4 ICR dataset. Furthermore, there were four States and three additional tribes or territories whose data are missing entirely from the analysis. A thorough QA/QC process was undertaken to evaluate these SYR 4 ICR data used for analyses. However, it is possible that data entry errors may still exist in the final SYR 4 ICR dataset. The QA/QC review focused only on the data elements essential for analysis as part of SYR 4. For a complete discussion of the SYR 4 ICR dataset, including a description of the quality assurance/quality control review, refer to the main text of this document and USEPA (2024a). For more detailed information on the microbial contaminants' occurrence analysis, refer to USEPA (2024b).

Section 10: Instructions on Importing SYR 4 Datasets

These text files are tab delimited and have no text qualifier. Field names are included in the first row of each file. The complete SYR 4 ICR dataset is too large to be imported into Excel as well as certain individual files, these files include individual years of TC and EC files, free chlorine, total chlorine and paired datasets of TC/EC/FC with residual disinfectant. The data are available for download for each parameter and should be imported into a data management system that supports large datasets for analysis.

10A: Downloading Data Files (Note that instructions may vary depending on the version and software used to import data.)

1. Begin by reviewing the SYR 4 ICR Dataset Summary (Exhibit E.2) and in particular note the table of Data Field Names and Definitions (Exhibit E.1).

2. Access the SYR 4 ICR data by going to the Six-Year Review homepage. Click on the link for "Six-Year Review 4."

3. Click on the desired zip file and select "Save As" to save the file to your computer.

4. Navigate to the location on your computer where you saved the zip file and extract the zip file contents by clicking "Open with" and using WinZip or a similar file compression software

10B: Importing Data into Microsoft Excel

Using Microsoft Excel 2013 or a newer version is recommended due to the size of the dataset(s). Note, the following microbial and disinfection byproduct data files are too large to import into Microsoft Excel: TTHM, HAA, free residual chlorine, total chlorine, all TC files, EC, and all paired microbes and disinfectant residual files.

5. Open a blank workbook in Microsoft Excel.

6. In the workbook, select Data among the tabs at the top of the page.

7. On the far left, top of the screen, go to the Get External Data section and select From Text.

8. You will be prompted to select a text file. Locate the text files you extracted in Step 4, and click "Import" on the text file of interest.

9. A preview of the file text converted to a table will appear. At the top, verify that File Origin (depending on your computer's operating system) displays "10000: Western European (Mac)" or "1252: Western European (Windows)." Select "Tab" as the Delimiter and "Based on first 200 rows" as the Data Type Detection. Click Load To...

10. In the next window, choose “Table” under Select how you want to view the data in your workbook. Select “Existing worksheet” for where to put the data and verify the table’s origin cell origin displays as “=A1.” Click OK.

11. A “Queries & Connections” window will appear on the right of the screen as Excel generates the new table. This step may take several minutes.

12. Save the Excel spreadsheet file once the table generation is complete.

10C: Importing Data into R

1. Open a blank R script.

2. Using the function read.delim(), import the text file using the following format:

    a. [analyte name] <- read.delim(file = [filepath], header = TRUE)

    Example: bromoform <- read.delim(file = "C:/Users/[username]/Desktop/SYR4-Microbes /SUMMARY_MDBPS_BROMOFORM.txt", header = TRUE)

3. Check the data frame that is generated to ensure correct formatting.

4. NOTE: data columns that should be in date format will be imported as character type. To fix the format, include the line "df$DATE <- as.Date.character(df$DATE, format = "%d-%b-%y")" in the R code, replacing df with the name of the dataframe, and DATE with the name of the column containing date information.

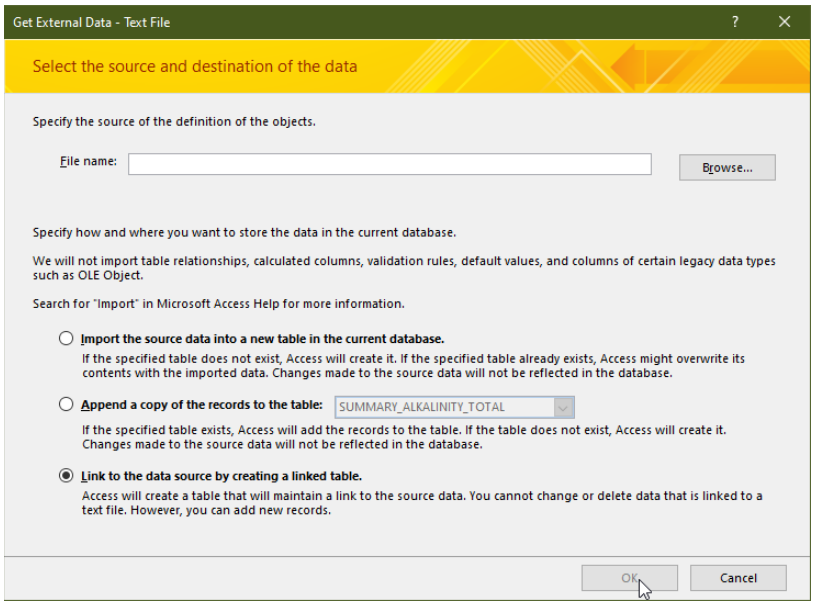
10D: Importing Data into Microsoft Access

1. Open a blank database in Microsoft Access.

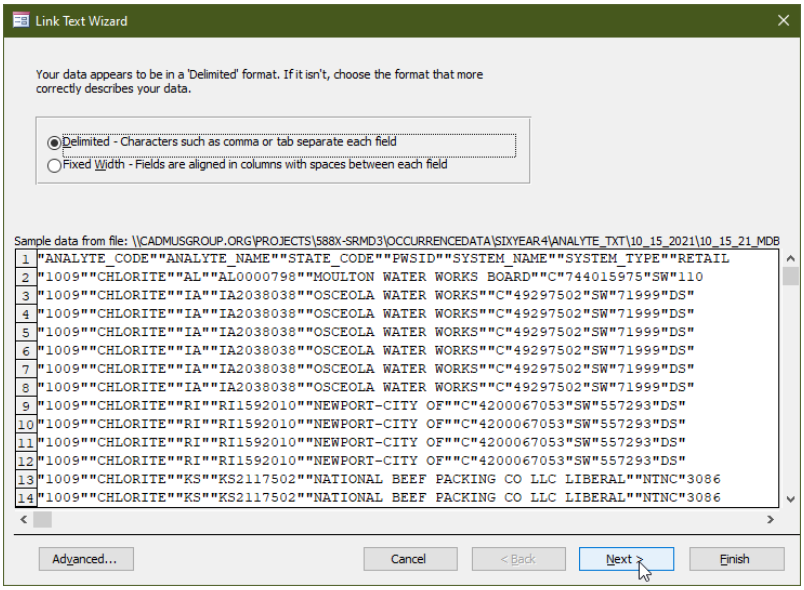
2. In the database, select External Data among the tabs at the top of the page.

3. On the far left, top of the screen, go to the New Data Source dropdown and select From File > Text File.

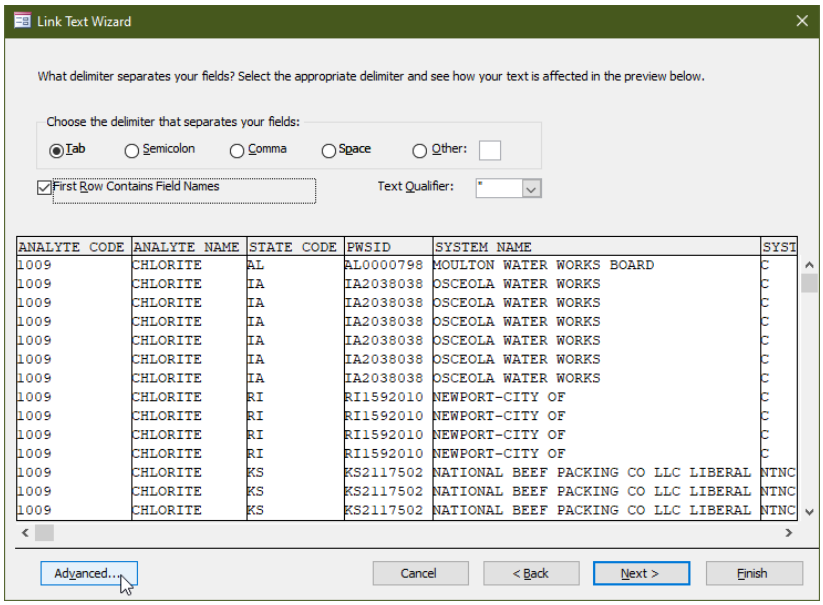
4. You will be prompted to select a text file. Locate the text files you extracted in Step 4, and with the following options: “import the source data into a new table in the current database”, or “Link to the data source by creating a linked table”. You can choose either method, but note that linking the file will maintain a smaller database size. Click OK.



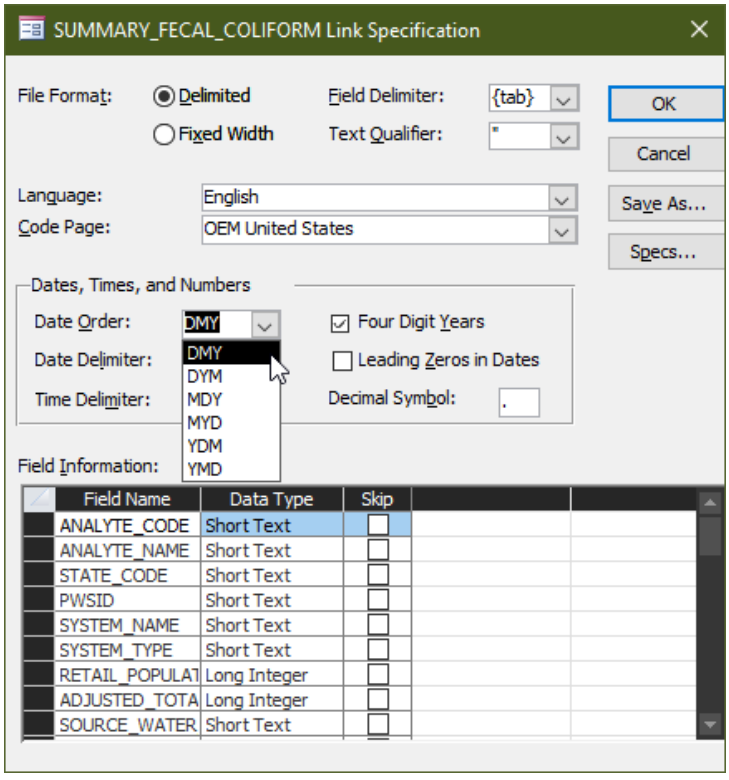
5. The Link (or Import) Text Wizard will appear. The default settings will be displayed and should have Delimited selected as the data format. Select Next>.



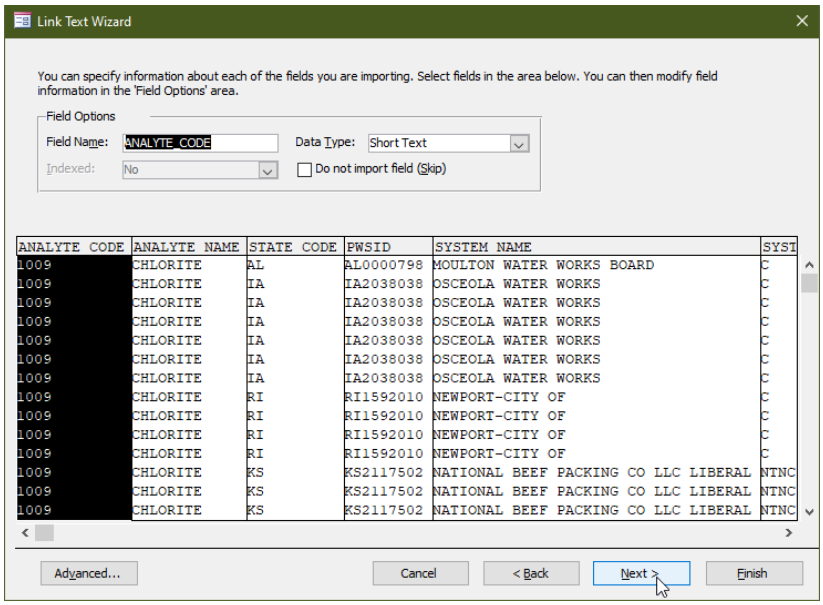
6. Default settings will display next and should have "Tab" selected as the delimiter. Select the checkmark box next to "First Row Contains Field Names." Next, click "Advanced…".



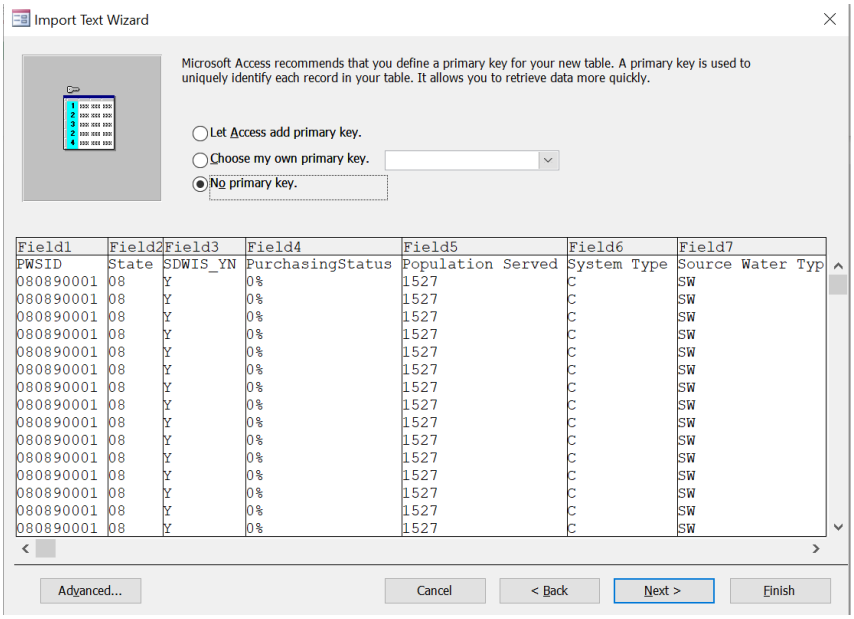
7. The Link (or Import) Specification window will appear. In the Dates, Times, and Numbers section, set the Date Order value to "DMY."



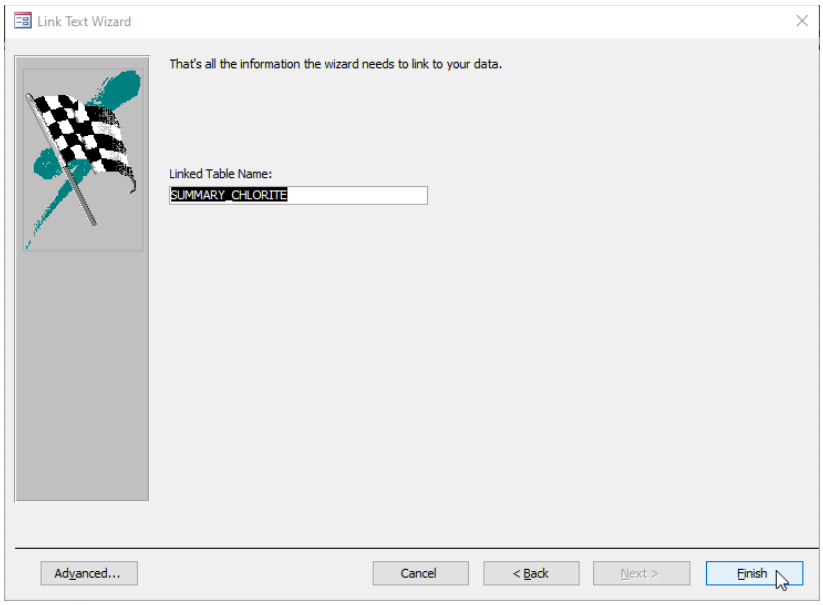
8. On the screen that follows, keep the default settings shown below and click Next>.



If you are importing instead of linking, a window will pop up related to setting a primary key. The default is set to "Let Access add a primary key". Check "No primary key" and click Next >.



9. A final screen will appear. Enter a meaningful name for the linked/imported table. This field will be auto-populated with the name of the linked file. Click Finish.



Part Two: Filtering and Formatting Data in Excel

10. To efficiently search, have cell A1 selected, choose "Data" among the tabs on the top of the page and click on "Filter." Each header title for each column now will have a small dropdown arrow displayed.

11. Filtering the data: a. If you want to look for a specific public water system, click the dropdown arrow for "PWSID" or "System Name." Within the search field, type the name and select from the displayed list. b. If you want to search for a different public water system, click the dropdown arrow and "Clear Filter from PWSID" or "Clear Filter from System Name." c. If you want to filter the data by contaminant, select "Analyte Name."

12. Multiple filters can be applied for example, allowing you to look for an individual water system's data for a specific contaminant of interest.

13. De-select Filter in the top menu bar and the entire database will again be displayed.

14. Note, all column formats are imported as the default General formatting. Column formats must be individually, manually changed in Excel after the download is complete to aid in data analysis. Use the Home screen in Excel, highlight the column and select the format from the drop-down menu. Suggested formats are:

Text fields	Analyte Name State Code PWSID System Name System Type Source Water Type Water Facility Type

	Sampling Point Type Source Type Code Sample Type Code Laboratory Assigned ID Sample Collection Date Detection Limit Unit Detection Limit Code Value Unit Presence Indicator Code
Numeric fields	Analyte ID Retail Population Served Adjusted Total Population Served Water Facility ID Sampling Point ID Six-Year ID Sample ID Detection Limit Value Detect Value Residual Field Free Chlorine mg/L Residual Field Total Chlorine mg/L

References

United States Environmental Protection Agency (USEPA). 2016. Six-Year Review 3 Technical Support Document for Disinfectants/Disinfection Byproducts Rules. EPA-810-R-16-012. December 2016.

USEPA. 2019. Information Collection Request Submitted to OMB for Review and Approval; Comment Request; Contaminant Occurrence Data in Support of the EPA's Fourth Six-Year Review of National Primary Drinking Water Regulations: October 31, 2019, Volume 84, Number 211, Page 58381-58382.